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Key factors and delivery mechanisms for passenger-oriented multimodal air journeys: A European practitioners' perspective

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ABSTRACT

Integrating air transport with alternative modes of transport holds great promise for substituting short-distance flights in regions like Europe, provided that the benefits of existing transport networks are preserved. Effective multimodal integration requires collaboration among transport operators and hubs to enhance the passenger experience. This study explored the perspectives of these practitioners across Europe through an interview-based case study, identifying key factors for a passenger-oriented multimodal air journey and examining the mechanisms that influence their delivery.

The findings reveal five Passenger Experience Factors (PEFs) that contribute to a successful passenger-oriented multimodal air journey: 1) integration of booking systems; 2) whole journey guidance; 3) transfer time and ease; 4) baggage management; and 5) disruption management.

We found that delivering these PEFs necessitates support processes within and between organizations to align, including IT, infrastructure, scheduling, operations, commercial, interorganizational, and strategic efforts. Additionally, external conditions – such as market dynamics and regulatory frameworks – play a pivotal role in either enabling or constraining these efforts. We discovered that these underlying support processes converge in delivery mechanisms that influence the provision of PEFs for multimodal air travel.

The findings emphasize that while airlines often lead in shaping the multimodal experience, progress is hindered by fragmented responsibilities, misaligned incentives among operators, and market competition. It is crucial to recognize that delivering passenger-oriented multimodal journeys requires effective cross-system collaboration, and that a regulatory framework must be established to create conditions for more sustainable transportation integration.

1. Introduction

Transforming airports into Multimodal Transport Hubs (MTHs) is presented as a promising strategy to reduce CO2 emissions (European Commission, 2021), alleviate airport congestion (Janic, 2011), and improve the competitive positioning of airports (Janic, 2011; Román & Martín, 2014). MTHs integrate both physical infrastructure and service elements of multiple modes of (long-distance) transport (Bell, 2019; Chauhan, Gupta, & Parida, 2021; Janic, 2011; Monzón, Hernández, & Di Ciommo, 2016; Toet, van Kuijk, & Santema, 2022).

The integration of air transport with alternative transport modes holds promise for regions like Europe, where the Trans-European Transport Network (TEN-T) is planned (European Union Agency for Railways, 2024; *Regulation (EU)* 2024/1679 on Union Guidelines for the Development of the Trans-European Transport Network, OJ L 2024/1679, 19.6.2024, 2024). High-speed rail (HSR) can emit up to 80% less CO_2 than air travel (Eurocities, 2024; European Union Agency for Railways, 2024). Buses are also more sustainable than flights for short

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distances, although this depends on the occupancy rate (Baumeister, 2019). Buses offer the additional benefit of significant operational flexibility in scheduling and network coverage (Vuchic, 2007).

While HSR and buses provide a more environmentally friendly alternative to short-distance flights, aviation retains advantages due to its extensive network and seamless connections to long-distance air travel (Li, Jiang, Wang, & Ma, 2018). Integrating sustainable transport modes with air travel through multimodal air journeys, while preserving the strengths of existing transport networks, can reduce airport congestion and enhance passenger experience and environmental performance (European Commission, 2021; Jiang, Wang, & Huang, 2022; Yang & Zhang, 2012).

Central to this endeavor is the transition within our economy from a focus on products to the development of purpose-driven systems, where interconnected services and user-centered design play a key role (Meroni & Sangiorgi, D., 2016). An example of this shift is the movement from uniform travel experiences to the design of multimodal journeys that emphasize the passenger experience (Camacho, Foth, Rakotonirainy, Rittenbruch, & Bunker, 2016; Canale, Tesoriere, & Campisi, 2019). Achieving such high-quality multimodal integration requires collaboration among the relevant transport organizations to provide a passenger-oriented experience (Docherty, Marsden, & Anable, 2018; Toet, Van Kuijk, Boersma, & Santema, 2023).

Despite increasing research on user-centered transport and multimodal travel, little is known about what transport practitioners consider essential for shaping multimodal air journeys around airport hubs. While existing academic studies primarily focus on the passenger perspective, understanding practitioners' viewpoints is important for two reasons. First, practitioners possess valuable knowledge shaped by their field experience (Malterud, 2001b). Second, their decisions and actions in real-life are influenced by their understanding of what constitutes a successful multimodal air journey. This means practitioners have crucial insights that can enhance our understanding of the mechanisms that enable or hinder the delivery of multimodal air journeys. This highlights the need for a qualitative study of practitioners' perspectives, as qualitative research is particularly well-suited to capturing the complexities of differing viewpoints and uncovering underlying challenges (Miles, Huberman, & Saldaña, 2014).

Through an interview-based case study, this study examines the perspectives of transport and hub operators across Europe and aims to contribute to the knowledge necessary for the successful delivery of passenger-oriented multimodal air journeys. The research is structured around two questions: *1) which passenger experience factors do practitioners consider essential for a successful multimodal air journey?*; and *2) What influences the effective delivery of these passenger experience factors to travelers in Europe?*

This study uses the Service Blueprinting framework (Bitner, Ostrom, & Morgan, 2008) as a lens to study the delivery of passenger-oriented multimodal air journeys. Service Blueprinting puts forward the notion that underlying support processes are essential for delivering user-centered services (Papastathopoulou & Hultink, 2012). It highlights the importance of a cross-functional approach in service creation, emphasizing that all organizational components must align towards the common goal of delivering coherent services. For multimodal air journeys, this alignment is crucial also between transport organizations, as multiple parties contribute to the creation of these integrated travel experiences (Toet et al., 2023).

The paper is organized as follows: Chapter 2 reviews the literature on transport systems and multimodal passenger experiences, while also outlining the theory of Service Blueprinting. Chapter 3 details the research methodology, and Chapter 4 presents the findings derived from the interviews. Finally, Chapter 5 discusses the implications and formulates conclusions.

2. Background

2.1. Transport networks

The transport system can be conceptualized as a network of interconnected transport subsystems, each fulfilling a specific function within the broader network (Aleta, Meloni, & Moreno, 2017; Geurs, La Paix, & Van Weperen, 2016; Kivela et al., 2014). An example of an interconnected transport system is the TEN-T project of the European Union (EU), which covers the main transport routes within the EU and exemplifies the integration of transport systems into a cohesive network, enabling seamless transitions between modes of travel (European Union Agency for Railways, 2024; Nechyporuk, Pavlikov, & Kritskiy, 2022; Regulation (EU) 2024/1679 on Union Guidelines for the Development of the Trans-European Transport Network, 2024). These transport systems include rail networks, aviation, buses, automobiles, and more, each dependent on its physical infrastructure, operational practices, and regulatory frameworks.

In the aviation sector, two main network structures are commonly used to organize transport. First the point-to-point network connects origin-destination pairs directly, making it attractive for direct routes. However, it struggles with traffic distribution, leading to imbalances on certain routes and capacity limitations, which increases overall costs due to the need to serve less profitable connections (Wang, Shen, Wang, Dong, & Li, 2024). The second network structure is the hub-and-spoke model, which uses larger, more significant hubs to serve smaller, non-hub nodes (Givoni & Banister, 2006; Wang et al., 2024). This model leads to fewer direct connections while allowing for more efficient use of the connections between hub and node airports, thereby enabling economies of scale on hub-to-hub routes (Wang et al., 2024). Disadvantages to this model are the additional capacity required for transfers at hubs, and the extra time needed for these transfers (Wang et al., 2024). In both types of networks, air transport passengers begin and end their journeys at node or hub airports, where they enter or exit the air system through airport procedures, including passport and security checks (Marquez, 2019).

Unlike air travel, rail networks are designed for intercity passenger transport and typically operate without security and passport checks. In Europe, the rail system evolved as a collection of national networks primarily focused on domestic travel (Bairras & Ardaiz, 2025; RLI, 2020), allowing passengers to board and disembark at multiple stations, often conveniently located in city centers (Vassallo & Fagan, 2007). However, rail transport is strongly dependent on physical infrastructure for its operations and for specific routes it follows. This reliance on infrastructure explains why building (high-speed) rail systems requires significant investment in both infrastructure and railway stations (Doomernik, 2015). While passengers frequently transfer between different train legs during their journeys, rail systems typically do not provide the same guarantees for missed connections as airlines do. This is because different legs of a long-distance international train journey are often operated by different companies, a legacy of their historical evolution as independent national networks (RLI, 2020).

Next, bus networks designed for longer distances generally operate on more flexible routes, often with designated stops, and are less reliant on physical infrastructure compared to rail transport. However, the reliability of these services is influenced by traffic conditions encountered along the route (Vuchic, 2007).

The TEN-T project places emphasis on HSR as a critical foundation

for sustainable multimodal transport (Nechyporuk et al., 2022). Airport hubs using the hub-and-spoke model can take advantage of this by incorporating trains but also buses as 'spokes' in their network, as long as the physical infrastructure of a modality is available near or preferably *at* the airport (Li et al., 2018). This integration enhances the connectivity and sustainability of the transport systems (Román & Martín, 2014).

2.2. Multimodal passenger experience

The growing emphasis on passenger experience and service-oriented approaches (Bitner et al., 2008; Meroni, & with Sangiorgi, D.)., 2016) has prompted research into the factors that define a seamless multimodal air journey (Camacho et al., 2016; Canale et al., 2019). The Mobility-as-a-Service (MaaS) concept is an example of multimodal transport in which mobility services across different providers are connected to enable a smooth passenger experience (Guidon, Wicki, Bernauer, & Axhausen, 2020). Previous research defines multimodal (air) journeys as those journeys that connect air travel with train or bus transport, with a single integrated ticket for seamless travel (Babić, Kalić, Janić, Dožić, & Kukić, 2022; Li et al., 2018; Román & Martín, 2014) and connection insurance to mitigate disruptions or missed connections (Román & Martín, 2014). Multimodal transport differs from intermodal transport in that it features a higher degree of integration among organizations (Babić et al., 2022; Rodrigue, 2024). Assessing the existing literature, we found that integrating multimodal air journeys requires attention to both service factors and physical infrastructure aspects (Bell, 2019; Chauhan et al., 2021; Monzón et al., 2016; Toet et al., 2022).

Several service-related factors influence the passenger experience. First, comprehensive journey information empowers travelers to plan journeys according to their preferences, while real-time flight updates enhance their perception of control (Allen, Bellizzi, Eboli, Forciniti, & Mazzulla, 2020; Antwi, Fan, Ihnatushchenko, Aboagye, & Xu, 2020; Cascajo, Lopez, Herrero, & Monzon, 2019; Hernandez Bueno, 2021; Watkins, Ferris, Borning, Rutherford, & Layton, 2011). Research suggests that communication of this information by staff further contributes to a positive travel experience (Antwi et al., 2020; Lubbe, Douglas, & Zambellis, 2011).

In addition to information provision, integrated ticketing simplifies the booking process by allowing a single purchase for an entire multimodal journey, thereby reducing complexity for passengers (Babić et al., 2024; Hickman, Chen, Chow, & Saxena, 2015; Román & Martín, 2014; Wang, Loo, & Li, 2020).

Moreover, minimizing friction during transfers is important in multimodal air travel. Key services include baggage integration (e.g., check-in at the departure station), dedicated multimodal touchpoints with trained staff to assist passengers, and transfer services (e.g., priority access at airport security) (Babić et al., 2022; Jiang et al., 2022; Li et al., 2018; Wang et al., 2020).

Last, these transfers should be coordinated to ensure smooth multimodal connections (Xia & Zhang, 2017). While short transfer times can induce stress, excessively long layovers may result in inefficiencies and inconvenience, negatively affecting the overall travel experience and attractiveness of product (Jiang et al., 2022; Román & Martín, 2014; Song, Hess, & Dekker, 2018). On the other hand, increased dwell time may benefit operators under certain circumstances (Xia & Zhang, 2017) and airports in terms of consumer spending (Wu et al., 2024).

From an infrastructure perspective, we identified three categories of factors that influence the passenger experience of multimodal air journeys. First, wayfinding and signage help travelers navigate efficiently through the transfer environment, making services easier to find and use (Allen et al., 2020; Nielsen, Eltved, Anderson, & Prato, 2021; Wang et al., 2020).

Second, the environment of a hub – characterized by shops, restaurants, well-maintained waiting areas, and safety and security measures – enhances the overall attractiveness of the transfer (Eboli & Mazzulla, 2015; Hernandez & Monzon, 2016; Hickman et al., 2015).

Third, the design of airport hubs, including terminal layout and walking conditions, significantly impacts the passenger experience (Geng, Yu, Lin, Wang, & Huang, 2017; Wattanacharoensil, Schuckert, & Graham, 2016). The train and bus stations should be located at or near the airport hubs, as proximity to these transport options is generally preferred (Li et al., 2018).

Finally, comfort during longer walks can be enhanced with amenities such as air conditioning and shelters (Hernandez & Monzon, 2016; Wang et al., 2020).

While existing research explores the passenger perspective, there is a gap in the literature regarding the viewpoints of transport practitioners. Comparing their perspectives with previously researched factors helps reveal misalignments between design and user needs. This highlights the need for further study on how multimodal collaborations are structured and implemented in practice.

2.3. Multimodality through service design

To make passenger-oriented multimodal travel a reality, actors from different transport systems must collaborate (Babić et al., 2022; Toet et al., 2022; Toet et al., 2023). And for airports to transition into MTHs, ecosystem-wide collaboration is required (Toet et al., 2023).

Li et al. (2018) describe that varying levels of collaboration exist among different transport operators, categorizing these partnership levels by elements such as frequent flyer programs, code-sharing agreements, integrated ticketing systems, schedule coordination, and end-to-end services. Previous research indicated that establishing such collaborations introduces challenges, as organizing large-scale alliances is difficult due to the differing interests of operators (Toet et al., 2023). Additionally, challenges related to investment responsibilities (Toet et al., 2023), revenue sharing (Babić et al., 2022), and responsibility sharing (Babić et al., 2022; European Parliament and of the Council, 2024) were identified. Furthermore, collaboration between transport operators for multimodality is hindered by data-sharing issues, as there are no common rules and standards between transport systems, resulting in data being collected and made available in different formats (Babić et al., 2022, Babić et al., 2024; Finger, Montero, & Serafimova, 2019). Li et al. (2018) emphasize the need to understand the driving forces behind these collaboration challenges in multimodal air travel.

As the passenger experience gains prominence, a holistic perspective, like service design, to the delivery of passenger-oriented multimodal air journeys, becomes increasingly relevant (Bitner et al., 2008; Mager, 2009). Applying a service design lens to the delivery of multimodal air travel enables studying the collaboration challenges between transport and hub operators that must be addressed.

One specific service design technique, Service Blueprinting, shows how the various service components are interconnected and maps out the characteristics of the service delivery process (Bitner et al., 2008; Shostack, 1984). This enables stakeholders to understand what the service entails and their respective roles in its delivery (Bitner et al., 2008). Service Blueprinting has been used in transport studies to redesign in-flight service processes (Nam, Ha, & Lee, 2018).

The method builds on customer journey mapping, which visually represents the sequence of customer interactions throughout a service

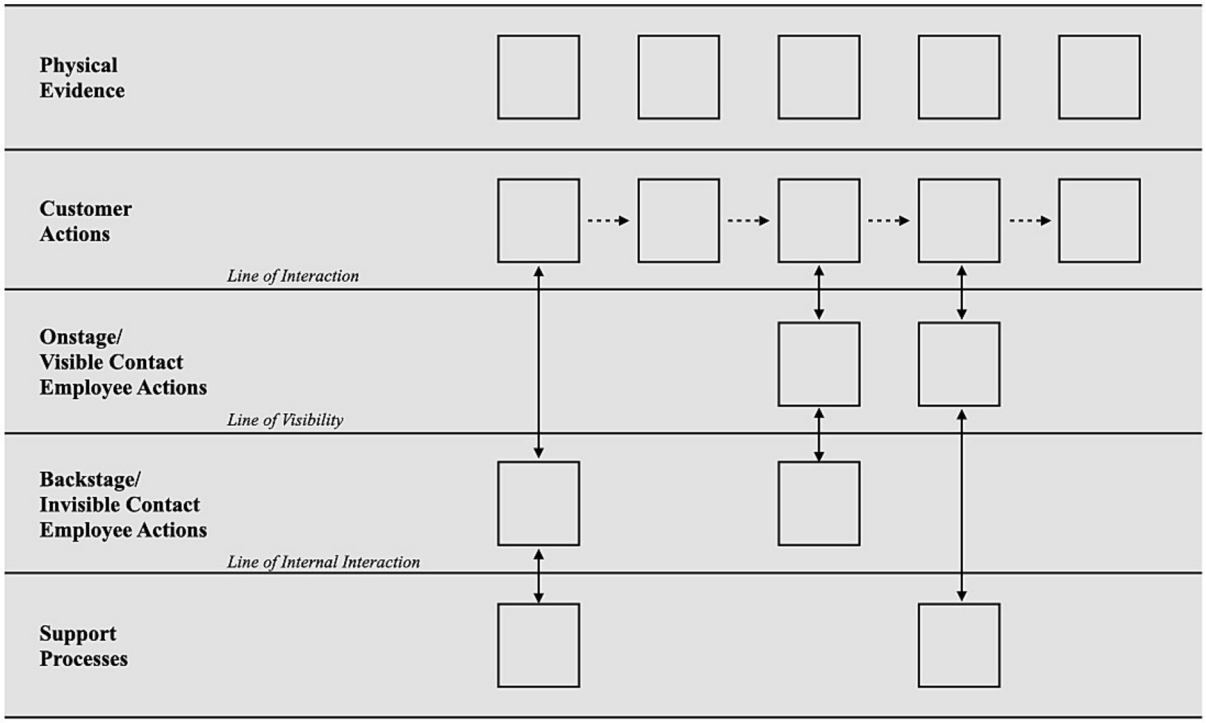

Fig. 1. Service Blueprint framework (Bitner et al., 2008).

experience (Rosenbaum, Otalora, & Ramírez, 2017). Service Blueprinting extends the customer journey map by adding three layers (Fig. 1): onstage/visible contact employee actions (interactions between customers and onstage employees or self-service technologies), backstage/invisible contact employee actions (interactions between customers and backstage employees), and support processes (underlying support processes not directly linked to customer interactions) (Bitner et al., 2008; Chakrabarti, 2019). The layers of the Service Blueprint are connected through vertical lines that indicate the interdependencies within the organization that ensure effective service delivery (Bitner et al., 2008). This makes the Service Blueprint a valuable framework for this study, which aims to identify how organizations deliver passenger-oriented multimodal air journeys.

3. Method

3.1. Qualitative, interview-based

To align with the exploratory nature of the study, a qualitative research approach was adopted (Miles et al., 2014; Patton, 2015; Silverman, 2009). This approach facilitated a deeper understanding of the complexities involved in developing multimodal air journeys, capturing data characterized by "richness and holism" (Miles et al., 2014). Data were collected through semi-structured interviews with key multimodal transport practitioners and analyzed using thematic analysis (Braun & Clarke, 2012). Interviewing is a suitable method because the purposeful selection and analysis of information-rich cases yield context-sensitive and in-depth insights that surpass the researcher's direct observational capabilities (Patton, 2015). Interviewing experts provides valuable insights into the domains in which each expert operates and highlights their influence within their respective fields (Döringer, 2021).

3.2. Case selection & informants

We conducted interviews with practitioners involved in delivering the three most advanced multimodal air&rail journeys offered through major European airport hubs, and in one contrasting case involving air&bus journeys. The cases were distributed across Western, Central, and Northern Europe.

This study purposefully focused on the European context (Miles et al., 2014; Patton, 2015; Silverman, 2009), where EU climate policies promote a shift from air travel to more sustainable modes of transportation (Chiambaretto & Decker, 2012; European Commission, 2011). The short intercity distances make trains and buses viable alternatives to short-haul flights (Schmitt & Gollnick, 2016).

Applying replication logic in a multiple-case study design, we selected cases within a bounded and comparable contextual setting to enable meaningful cross-case comparison and support analytic generalization (Yin, 2018). Therefore, regions outside the EU were excluded due other contextual settings, such as limited HSR infrastructure, reliance on competitive transport rather than integrated approaches, and differing governance systems (Nash, 2010; UIC, 2024; Veeneman, 2023; Wan, Ha, Yoshida, & Zhang, 2016; Zhou, Tan, & Sedlin, 2018).

Through an initial scan, based on desk research and inquiries among

Table 1
Inclusion criteria for the selection of airport hubs.

Inclusion criteria for the selection of Airport hubs
1) The airport hub features an integrated train or bus station, as literature highlights the importance of proximity between transport modes in facilitating passenger-oriented multimodal connectivity (Li et al., 2018).
2) The airport hub operates within a hub-and-spoke network, as hub airports were considered to have the greatest potential for integrating long-distance travel with other (short and medium distance) transport modes.
3) The airport hub is located within in the EU to align to ensure similarity in regulations.
4) The airport hub handles over 20 million passengers in 2019 to ensure comparable levels of operational complexity, including security processes.

Table 2
Inclusion criteria for journey selection.

	Initial selection					
	Final selection					
Journey inclusion criteria (all must be met)	**Case 1**	**Case 2**	**Case 3**	**Case 4**	**Case 5**	**Case 6**
1) Journey consisted of an air leg and one leg of train or bus transport (*Toet* et al., *submitted*).	*Train*	*Train*	*Train*	*Bus*	*Train*	*Train*
2) Journey was offered via a single integrated ticket (Babić et al., 2022; Li et al., 2018; Román & Martín, 2014).	*V*	*V*	*V*	*V*	*V*	*V*
3) Provides connection insurance for travelers (Román & Martín, 2014).	*V*	*V*	*V*	*V*	*V*	*V*
Service inclusion criteria (at least one must be met)						
4) Offers comprehensive baggage handling for multimodal air travelers.		*V*			*V*	
5) Dedicated multimodal touchpoints with trained staff.	*V*	*V*	*V*			
6) Makes use of specialized transfer services, such as priority at airport security.	*V*		*V*	*V*		

airport professionals, we identified eight European airports where to some extent multimodal air journeys were being offered. We then applied predefined inclusion criteria to the airport properties to ensure a certain level of airport complexity and case comparability (see Table 1) (Yin, 2018).

Next, we assessed the multimodal air journeys available at the initial selection of airport hubs. Journeys were only considered as candidate cases if they met the three journey criteria and at least one of the service criteria. Thus, we identified six potential cases (Table 2).

Out of these, we selected four cases for investigation. Cases 1, 2, and 3 because these were the most advanced; these cases adhered to 1) all journey inclusion criteria in Table 2, as well as to 2) at least two of the service inclusion criteria in Table 2. They also were comparable (Miles et al., 2014) in terms of journey characteristics and service offerings. The study's resources provided capacity to investigate one additional case, and in addition to the three most advanced and similar cases, we selected Case 4, which featured a bus as the ground transport mode, as an outlier case (Miles et al., 2014). Including an outlier case is valuable, as it can enhance the robustness of the findings when patterns are consistently observed in the comparable cases but not in the contrasting case (Miles et al., 2014).

Data were collected through semi-structured interviews with key transport practitioners actively involved in delivering passenger-oriented multimodal air journeys at four European airport hubs. Following a purposive informant sampling strategy (Miles et al., 2014; Patton, 2015) for each case, the key representatives from airlines, airports, and rail operators were selected and interviewed (see Table 3). During the recruitment of the informants, we found that Case 1 involved two rail practitioners sharing responsibilities for multimodal operations, whereas Case 4 included only airline and airport practitioners, as the bus service was operated by the airline. For confidentiality reasons, the names of the cases and practitioners are not disclosed.

These interviews represent a carefully targeted selection of key individuals directly responsible for the development and delivery of the multimodal air journeys studied. Since such journeys are still relatively rare in practice, it is not always publicly clear which individuals are involved. Identifying and reaching the right practitioners therefore required considerable time and effort. This focused approach enabled the study to capture rich, experience-based insights from the most relevant experts in the field. Furthermore, the total number of interviewees ($N = 12$) is within the empirically supported range of 9 to 17 interviews for achieving saturation in qualitative research (Hennink & Kaiser, 2022).

Table 3
Overview of the informants who were part of the interview study, by their type of company and case.

Case	ID	Company type
Case 1 (Air&Rail journey)	Airline, C1	Airline operator
	Airport, C1	Airport infrastructure manager
	Train1, C1	Train operator
	Train2, C1	Train operator
Case 2 (Air&Rail journey)	Airline, C2	Airline operator
	Airport, C2	Airport infrastructure manager
	Train, C2	Train operator
Case 3 (Air&Rail journey)	Airline, C3	Airline operator
	Airport, C3	Airport infrastructure manager
	Train, C3	Train operator
Case 4 (Air&Bus journey)	Airline, C4	Airline operator
	Airport, C4	Airport infrastructure manager

3.3. Data collection

The semi-structured interviews utilized a general interview guide to ensure consistency among informants (Patton, 2015). The guide focused on two primary topics:

1. The passenger experience of multimodal air travel: informants' perspectives on passenger-oriented multimodal air services, including strengths, areas for improvement, and rationales for current practices; and
2. The delivery of passenger-oriented multimodal air travel: the service's delivery process, critical success factors, challenges, and future enhancement plans.

The interviews were conducted between February and October 2023 by the first author via Microsoft Teams. Sessions were recorded and transcribed. The interviews were conducted in English or in the shared native language of the interviewees. The duration varied from 40 to 60 min.

3.4. Data analysis

The interview transcriptions were thematically analyzed using the six phases established by Braun and Clarke (2006): (1) familiarizing with the data; (2) generating initial codes; (3) searching for themes; (4) reviewing themes; (5) defining and naming themes; and (6) reporting findings.

Data familiarization involved listening to the recorded interviews, transcribing them verbatim, and reviewing the resulting transcripts. These transcripts were the primary data source for subsequent steps.

The coding phase was performed by using the Goia coding technique, and was conducted in iterative rounds (Magnani & Gioia, 2023). First, the primary author created first-order codes, in which the codes were represented by informants' understandings. This was done using the ATLAS.ti qualitative data analysis software. The first-order codes were reviewed by the other authors by using an analytical framework based on Service Blueprint theory (Fig. 2) (Bitner et al., 2008), as a lens for interpreting the data (Magnani & Gioia, 2023; Malterud, 2001a).

The review of the first-order codes facilitated the creation of second-order codes (the researcher-based understandings), and led to the division of two main groups (Magnani & Gioia, 2023): Passenger Experience Factors (PEFs) (designated as A in Fig. 2) and delivery mechanisms (letter B). The PEFs encompass codes that align with the four top layers of the Service Blueprint model – physical evidence, customer actions,

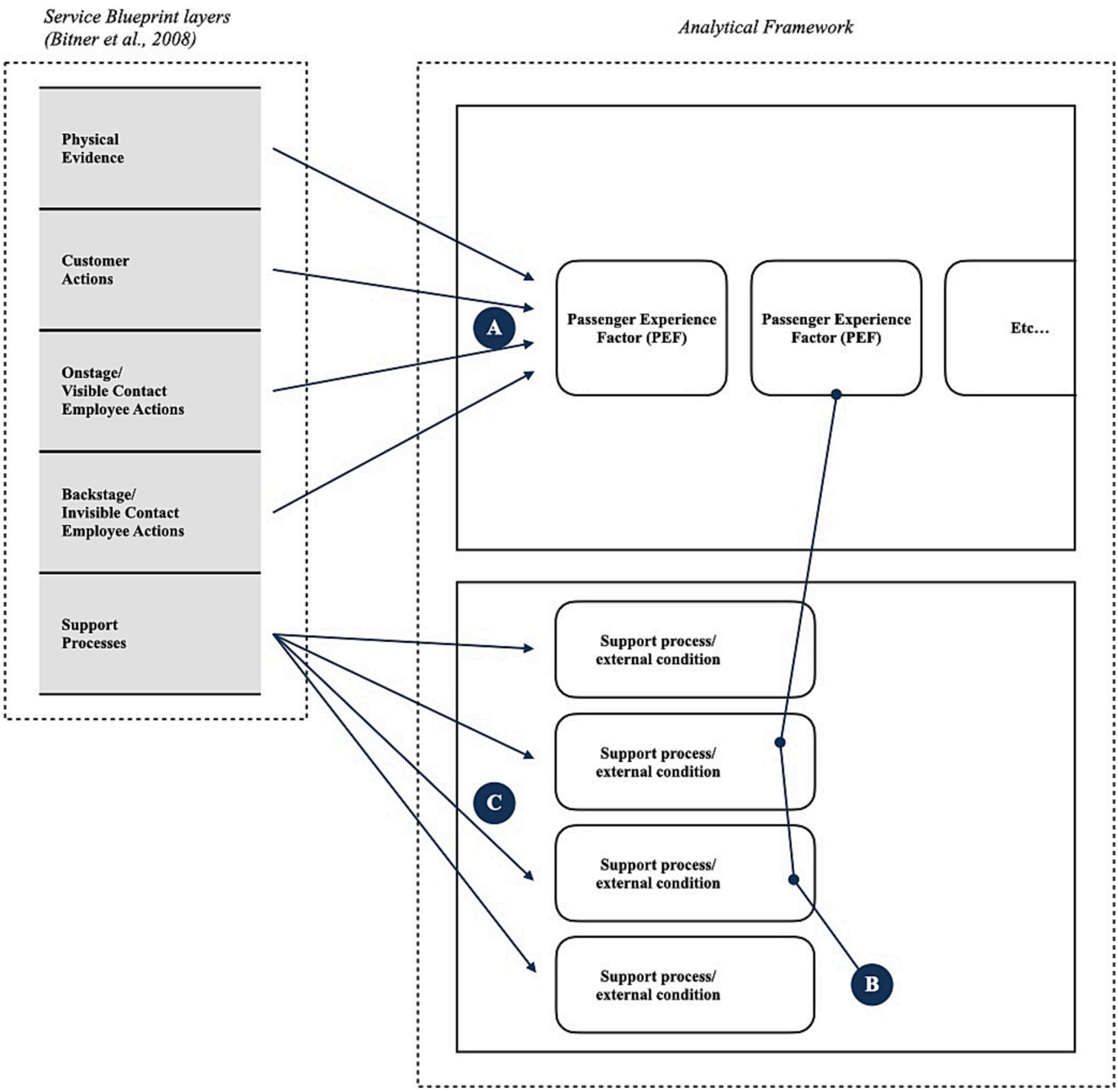

Fig. 2. Analytical Framework based on the Service Blueprinting theory of Bitner et al. (2008).

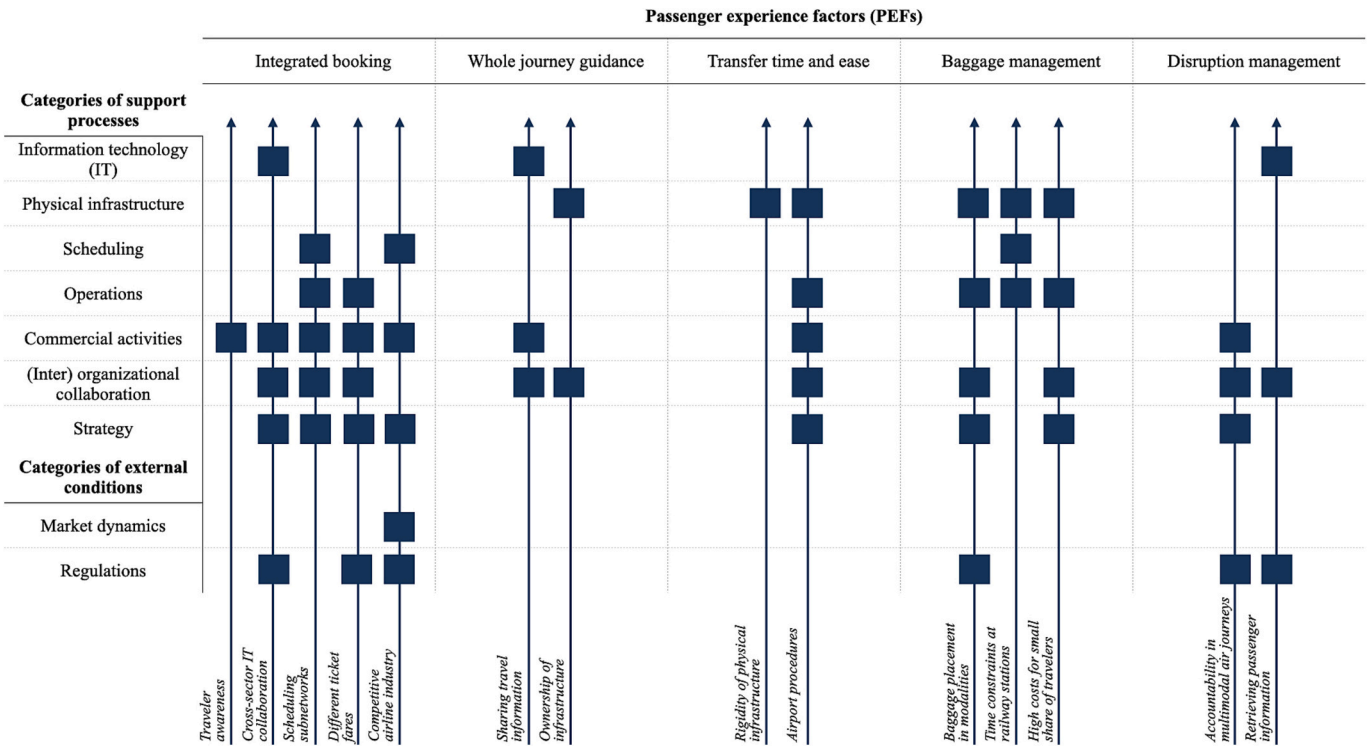

Fig. 3. An overview of how the delivery of PEFs in multimodal air journeys is driven by mechanisms of support process and external conditions.

onstage actions, and backstage actions – directly relating to the passenger experience. The delivery mechanisms illustrate how the PEFs are delivered through support processes, consistent with Service Blueprint theory (Bitner et al., 2008).

During the data analysis, we found that the delivery mechanisms consist of multiple categories of support processes and external conditions (letter C).

The first author identified the PEFs, mechanisms, and categories of support processes and external conditions, which were reviewed by all authors to ensure consistency and validity. Illustrative quotes from informants were collected and paraphrased for clarity (and translated if necessary). Finally, all authors reviewed the report.

4. Results

This chapter presents findings on PEFs for multimodal air journeys and their delivery from practitioners' perspectives. The study identified five key PEFs as core objectives for successful multimodal air journeys: 1) integrated booking, 2) whole journey guidance, 3) transfer time and ease, 4) baggage management, and 5) disruption management. For each PEF, we found mechanisms that influence its effective delivery. An overview of these findings is presented in Fig. 3. The top row displays the PEFs, and the vertical lines represent the delivery mechanisms and indicate the interdependencies between the support processes and external conditions. The categorization of the support processes is provided in Table 4, along with a definition of each category.

The remainder of this chapter concentrates on describing each PEF, followed by an explanation of the delivery mechanisms for each PEF.

4.1. PEF: integrated booking

The informants underscored that integrated booking is essential for the successful execution of multimodal air journeys. By consolidating various travel segments into a single itinerary, integrated booking simplifies the planning of these journeys. Key components of integrated booking include a single (digital) ticket that encompasses all modalities, a streamlined booking procedure, and reduced costs compared to separate tickets.

Table 4
The categories of support processes and external conditions, along with working definitions.

Categories of support processes	*Definition*
Information Technology	Use of computers, software and networks to create, manage and deliver transport. It enables the storage, processing and communication of data.
Physical infrastructure	Physical assets that support transport processes, such as roads, rails, stations and terminals (Toering, De Bruijne, & Veeneman, 2025).
Scheduling	Planning and timing of transport services and modalities.
Operations	Day-to-day activities required for the coordination of resources, processes, and personnel to ensure smooth execution of transport.
Commercial	Activities focused on business viability and value creation for the organization(s) involved, including efforts to market services, sell tickets and promote awareness of multimodal travel options.
(Inter)organizational	The internal structures and processes, and collaboration among actors, that enables an organization to function effectively.
Strategy	A high-level plan that defines the organization's goals and actions to achieve them.
Categories of external conditions	*Definition*
Market dynamics	Dynamics that influence the behavior of market actors, and covers aspects such as competition, consumer demand, pricing trends, market entry barriers, and evolving business models.
Regulations	The policies, legislation and regulations set by government bodies that affect the delivery of transport.

A significant advantage of integrated booking is the capacity to search for and reserve all trip segments through a single online platform. Informants highlighted the importance of eliminating the need for passengers to navigate multiple websites: "*I think this is the most important step in order to show the customer, OK, there is an option with the rail as a first or last segment.*" (Train, C3).

In addition to enhancing convenience, a single ticket functions as a unified contract for the entire journey, also providing financial benefits. Informants noted that integrated booking frequently offers discounts or bundled pricing compared to the purchase of separate tickets.

The research found that five mechanisms shape the delivery of integrated booking, which are outlined below.

4.1.1. Traveler awareness

Passenger awareness is a requirement in the adoption of integrated booking for multimodal air journeys. Informants highlighted that many travelers are unaware of the option to book these journeys as a single itinerary, which leads them to purchase separate tickets for train and air segments. Traditionally, trips are booked airport-to-airport, while multimodal options include connections from nearby rail stations. However, this lack of awareness about these possibilities prevents passengers from taking advantage of integrated tickets: *"The huge majority of clients buy a train ticket and then a plane ticket separately. And so, they take the risk of missing the connection."* (Train, C2).

Without sufficient knowledge about the option to book an integrated ticket, the demand for such services remains low, which in turn reduces the effectiveness of offering multimodal air journeys in the first place.

4.1.2. Cross-sector IT collaboration

Establishing integrated digital ticketing requires close collaboration between the air and rail sectors, but differences in their IT structures create challenges.

Airlines rely on standardized IT systems for information and code sharing, which are not commonly used in the rail industry. Our findings suggests that rail operators often adapt to airline standards to enable integrated booking: *"We align ourselves to the airline standard in order to sell our trains as flights. And I think this is the basic thing to sell this because the airlines are not willing to align to our standards, so therefore the railway companies need to align to the airline."* (Train, C3).

While airlines do have extensive experience collaborating within their own system, extending this cooperation to other transport systems to support integrated booking, requires significant adjustments in existing IT structures.

4.1.3. Scheduling subnetworks

The scheduling properties of different transport subnetworks do not align efficiently to enable integrated booking for all types of multimodal air journeys.

Informants noted that parts of the European air transport network, primarily those parts involving intercontinental travel, rely on early morning feeder flights to ensure timely connections with the western part of the world. To achieve these worldwide connections, the air travel industry has a long history of international cooperation among various air operators. In contrast, the rail sector has developed as national networks, transporting passengers from city to city. While trains in Europe can be efficient, travel times are often extended due to multiple stops along the way. Therefore, replacing feeder flights with buses or trains presents significant scheduling challenges: *"The first planes are taking off very early in the morning and the last ones are coming back very late. So if you come back with those planes, you don't have any train afterwards to go back home."* (Airport, C2).

Operating trains early in the morning or late at night would require increased rail service frequencies. However, this is constrained by the rail system's reliance on nighttime hours for maintenance, as one rail operator explained: *"Our trains can't run at night because we do maintenance, and if you want to cover all the hub sections, then we should have trains very early in the morning. So, departing very, very early from outbound stations or very late, it is technically impossible for us."* (Train, C2). Notably, informants indicated that rail operators have little incentive to run trains more frequently to connect with air transport, as it is not economically attractive for them as the numbers of transfer passengers are not high enough to justify running solely for that purpose.

4.1.4. Different ticket fares

Interviewees noted that variations in ticket fares across transport modalities present a challenge for airlines in implementing integrated booking for multimodal air journeys. In hub-and-spoke networks, a short-leg flight that is part of a multi-segment air journey with a hub transfer may be priced lower than the same flight when booked as a standalone point-to-point connection. To prevent fare abuse and protect revenue integrity, airlines enforce boarding checks. However, this becomes more complex when a single booking includes both a flight and a train segment, as rail systems typically do not have the same stringent boarding controls as airlines. As one airline representative explained: *"For example, for competitive reasons, the ticket from Amsterdam to New York via Frankfurt airport could be cheaper than the ticket from Frankfurt airport to New York. And if someone would be boarding the train or would pretend to board a train in Amsterdam just to get a cheaper fare out of the hub Frankfurt airport, yeah, that would be a problem."* (Airline, C3).

To address this issue, informants from Case 1 referred to the Air&Rail desk at Brussels South train station, where multimodal travelers must collect their boarding passes to ensure compliance with the full journey: *"If the passenger does not pick up his boarding pass at the [Air&Rail] desk, well, persons cannot board the train and cannot use the plane."* (Train 2, C1).

4.1.5. Competitive airline industry

The rise of multimodal air journeys is largely driven by airlines responding to sustainability pressures. Facing criticism for operating short-distance flights, airlines have started collaborating with other transport providers to offer integrated travel options. However, competition within the airline industry ensures that traditional flights will continue to coexist with these multimodal alternatives.

This competitive dynamic creates a disincentive for airlines to fully commit to integrated booking for multimodal air journeys. By offering such options, airlines risk losing passengers – and consequently revenue – to competitors operating through other hubs. One airline informant explained that airlines may even continue operating certain short-distance routes, despite the availability of multimodal alternatives, in order to protect their flight slots: *"If you don't have any other use for the [flight] slot, you may keep flying on a certain route because you want to protect the slot."* (Airline, C3).

According to informants, the absence of European or global legislation reinforces this situation, allowing airlines to prioritize competitive interests over fully depending on multimodal travel.

It is the case that when airlines recognize strategic value in offering multimodal options, they become more inclined to invest in seamless booking systems. As one informant observed, growing attention to multimodal transport allows airlines to extend their networks into regions that were previously unserved by air travel: *"It allows [the airline] to drain more customers by the train, let's say in the 300/400 kilometers around [the airport], where anyway you wouldn't have any aircraft."* (Airline, C2).

4.2. PEF: Whole journey guidance

Clear passenger guidance is essential for a smooth multimodal air journey, both in pre-travel preparation and in providing real-time assistance during the journey. While this section focuses on overall journey guidance, transfer-specific considerations are discussed in *PEF 3: Transfer time and ease*.

Informants emphasized the need for clear multimodal instructions at

the booking stage to ensure that passengers understand their journey details, including their rights and responsibilities. They explained that travelers may be unaware that they have booked a multimodal trip, leading to confusion upon arrival at a rail station. As one informant noted, *"A lot of those people simply don't realize that part of the journey is by train, so we really need to explain that better"* (Airline, C1).

Beyond pre-travel communication, clear wayfinding, baggage handling instructions, and real-time updates are essential, especially at airports and train stations. Informants stressed the need for reliable guidance throughout the journey to minimize uncertainty and improve the overall passenger experience. The importance of both digital and physical support tools, such as instructional videos, was explained: *"That instruction video is powerful; I think it is an essential component to help reassure passengers"* (Airline, C1).

These measures collectively contribute to a well-coordinated travel experience, reducing uncertainty and enhancing passenger confidence in multimodal air journeys. This study identified two delivery mechanisms that shape the provision of whole journey guidance services, as detailed below.

4.2.1. Sharing travel information

Effective passenger guidance in multimodal air journeys relies on information sharing between transport systems, which in turn depends on IT-system integration. However, technical limitations currently hinder this exchange. One airline informant highlighted that rail data cannot be directly incorporated into airline systems, requiring workarounds that complicate the process: *"Setting up communication between especially the rail partners and airlines is not an easy business. Airlines have been developing standards for inter-airline communication for 30 to 40 years. There's nothing like that on the rail side for example." (*Airline, C3).

4.2.2. Ownership of infrastructure

The lack of control over digital and physical infrastructure limits efforts to improve passenger guidance. For example, while transport operators strive to enhance wayfinding, signage and directional information are managed by station and airport authorities. This limits operators in their ability to implement changes:*"Most of the station [owners] don't like when the partners are playing with signage and everything, so they want to - and I can understand - have it the way they want."* (Train2, C1).

4.3. PEF: transfer time and ease

Informants identified the transfer as the most critical juncture in multimodal air journeys, emphasizing two key aspects: time and ease.

First, efficiency is crucial for multimodal air transfers, as passengers need sufficient time for security checks, baggage handling, and boarding, while remaining competitive with air-to-air connections. Longer transfer times may deter passengers who prefer air-only travel. Physical proximity between transport modes shortens distances, reduces the needed transfer times, and makes the journey more appealing to passengers.

Second, a passenger-oriented multimodal experience relies on transfers that make passengers feel at ease. Informants emphasized that this can be achieved through well-integrated touchpoints, such as check-in desks and baggage drop-offs, as well as through clear and intuitive wayfinding. Informants highlighted that air-only transfers generally offer better amenities, especially in waiting areas, compared to multimodal transfers. Train stations were often noted to lack the comfort found in airports, emphasized by this informant: *"You're at [the train station], and your train leaves in 1.5 hours. Where are you going to wait? At [the airport], you can comfortably wait at the gates with benches and all that. Here, there's nothing" (Airline, C1)*.

Combined, these elements enhance the overall passenger experience. Below, we outline two delivery mechanisms identified in this study that influence these aspects.

4.3.1. The rigidity of physical infrastructure

Stations and airports were not originally designed with multimodal transport in mind, complicating the smooth transfer of passengers between different modes of transport. As one informant explained, *"This problem with the old infrastructure [...], they were not designed with the idea that one day you have to put some desk of air companies."* (Airport, C2). This results in long walking distances between different transport modes and complex station layouts, making improvements particularly challenging in terms of timelines and costs.

4.3.2. Airport procedures

Airport procedures, such as customs, security checks, and baggage handling, present significant hurdles for multimodal air travel. These processes, typically confined to only the departure airport in air-only travel, must be performed during the transfer stage of multimodal air journeys – such as when passengers transfer from a train to a flight. This increases the time needed for transfers.

A potential solution suggested by an informant is to provide priority services during airport transfers to help streamline the process: *"One key thing for us is to provide a smooth connection throughout the terminal and security at the [airport]. So that's why we are offering priority services whenever you are catching a flight"* (Airline, C4). However, not all informants agreed with this solution, particularly those from the airline sector, noting that priority services are typically reserved for loyalty (high yield) members.

Another suggested solution for successful multimodal air transfers includes the creation of a dedicated transfer filter at airports, designed to facilitate smoother transitions between modes of transport. However, such a solution would require significant investment from airports.

4.4. PEF: baggage management

The data highlighted baggage management as a key challenge in multimodal air journeys, with differing perspectives on how it should be handled. While some informants emphasized the importance of streamlining the baggage process throughout the entire journey, they argued that full integration across transport modes is not essential. Instead, they proposed practical solutions such as dedicated luggage racks on trains and specialized baggage handling at airports, all aimed at reducing the inconvenience of carrying luggage. For passengers facing long walking distances in terminals, a quick baggage drop-off after disembarking from the train was seen as particularly beneficial: *"we had some baggage drop off and delivery very close to the train platform. That is a strength that is very important for [the airport]."* (Airline, C3).

Others, however, advocated for a fully integrated baggage system, as is the case in air-to-air travel, where passengers would not need to handle their luggage between transport modes. They pointed to baggage service companies that already facilitate seamless transfers by offering check-in at train stations, allowing passengers to travel without the burden of managing their bags.

Despite differing opinions on the ideal approach, there was broad consensus on one key point: full integration without third-party involvement was generally considered unrealistic. The data revealed three underlying mechanisms contributing to this view, with one key message being that baggage handling in multimodal air travel is often expected to meet the same standards as in air-to-air travel. This expectation, particularly expressed by informants from airlines, is seen as a significant barrier to achieving full multimodal air integration.

4.4.1. Baggage placement in modalities

A key challenge lies in determining where to place baggage on trains. While aviation security regulations mandate a controlled environment for checked baggage, informants pointed out that European trains are not designed to accommodate large volumes of luggage. As one informant explained: *"Did you ever realize that European trains don't tend to have our luggage cars anymore? And the air industry authorities require a*

controlled environment for checked baggage." (Airline, C3). This highlights a significant issue in trying to integrate baggage management across different transport modes.

4.4.2. Time constraints at railway stations

Unlike planes at airports, where baggage handling is a well-established process, or buses at stations, trains are required to stop briefly at stations, making it nearly impossible to load or unload large volumes of baggage in such short durations. One informant explained: *"If you want to copy-paste the current experience passengers have had for decades [at the airport], that would mean that when the train arrives in the airport railway station, you have some handling of the bag – since the passenger wouldn't take care of this – that would pull the bag out of the train in probably 2 minutes. That's pretty challenging if you don't want to change the railway network […], I don't think it's achievable."* (Airline, C2). Buses, on the other hand, offer greater flexibility as they are not bound to fixed tracks and can wait for delayed passengers when necessary.

4.4.3. High costs for small share of travelers

Implementing full baggage integration would require significant financial investment and additional personnel at multiple locations, and affecting the current rail network. However, multimodal air travel constitutes only a small part of the rail industry, making it challenging for operators to justify investments in dedicated multimodal touchpoints at stations and airports. One informant highlighted this issue: *"And then the nightmare of having the facilities set up at all [stations]. That all would have to be manned, that all would have to be staffed, would be extremely cost-intensive, and the space is not there."* (Airline, C3). This is further complicated by the involvement of large, established organizations, meaning that the employees of these organizations must also be persuaded to invest in multimodality.

4.5. PEF: disruption management

Disruption management emerged as a crucial factor in ensuring passengers are well cared for during disruptions or are empowered to handle them independently, promoting both reliability and a sense of security. This aspect is closely linked to 'PEF 1: Integrated booking' and 'PEF 2: Whole journey guidance,' as passengers need to be kept informed about changes, potential actions, and their rights in the event of disruptions.

While disruption management is a standard practice in air travel, handling disruptions in integrated multimodal air journeys requires special attention. Interviewees stressed that passengers must be guaranteed connections and rebooking onto alternative transport options in case of missed transfers: *"That passengers on the train can be sure that they reach their flight and, if not, they will be rebooked and all that stuff. So that is taken care of"* (Airport, C3).

As one informant pointed out, passengers tend to prioritize assurance over speed. For long-distance travelers in particular, the security of knowing their needs will be addressed during disruptions was mentioned as more important than simply minimizing travel time: *"Mainly for long [distance] travelers, I think that they are not looking for the fastest option. I think they need to be as assured as possible"* (Airport, C2).

The findings reveal two key delivery mechanisms that play a role in disruption management, which are discussed below.

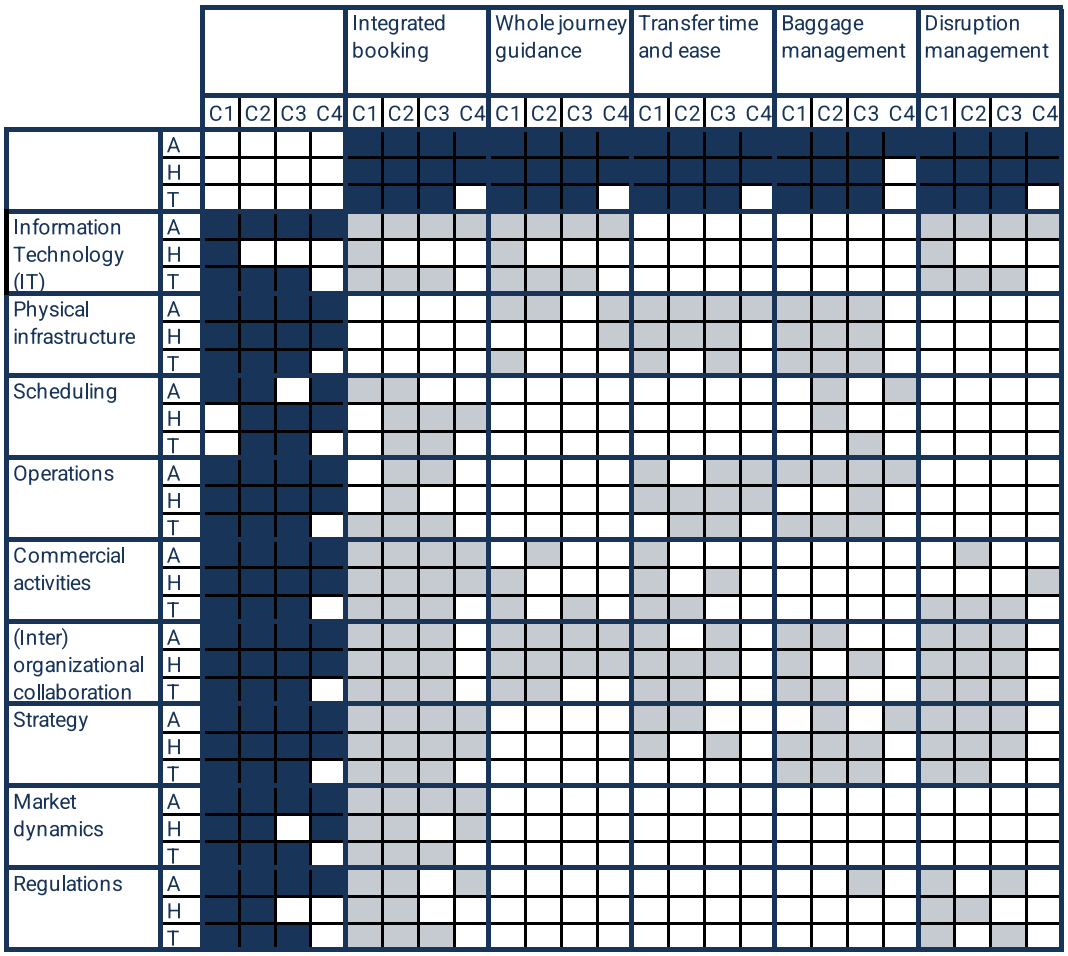

Fig. 4. An overview of the presence of PEFs and the categories of support processes and external conditions in the data.

4.5.1. Accountability in multimodal air journeys

It was found that there are no clear rules regarding the ownership of multimodal air journeys, which is detrimental to the passenger in case of disruptions. One airline representative pointed out the problem: *"Normally, in a codeshare, the operating carrier causing the issue is also responsible for resolving it and compensating passengers. [...] But in this case, no one is responsible, and that simply doesn't work. You can't accept passengers on such intermodal connections without clear accountability. That's something we really need to address in Europe through legislation."* (Airline, C1). Consequently, the informants noted that in their cases, the airlines assume responsibility for managing disruptions, as this aligns with their role as the ticket vendor.

4.5.2. Retrieving passenger information

Transport operators need to have accurate data on who is on board a vehicle, so they can respond appropriately and communicate with passengers affected by disruptions. As one informant explained, *"We brought in the IT teams to ensure everything was really robust in terms of the booking systems, so we know exactly who is on our train, and in managing disruption services. So now, we've taken a few steps to operate more efficiently"* (Train 2, C1). Beyond the technical challenges of different IT systems, regulatory frameworks also influence the feasibility of system integration. Establishing partnerships is essential for facilitating information exchange, but government regulations can complicate collaboration, especially for airports that hold a monopoly. As one informant pointed out, restrictive policies prevent airports from forming agreements with individual transport operators: *"Because of this monopoly situation and the agreements directly between [airline operator] and [rail operator], it's hard for us at the airport to get some data regarding even how many people are using this service"* (Airport, C2). Consequently, having a single operator manage the journey can complicate effective disruption management among all involved parties.

4.6. Groundedness of the findings

This section evaluates the extent to which the data supports each PEF and the categories of support processes and external conditions in the delivery of passenger-oriented multimodal air journeys (Eisenhardt, 1989). Fig. 4 visualizes these connections: the first row indicates whether a PEF was mentioned by cases and informants (black) or not (white), while the first column shows mentions of each support category, providing insight into their perceived importance. The subsequent rows and columns illustrate the presence (dark grey) or absence (white) of relationships, as well as the cases and informants mentioning these connections.

Examining the first row reveals that all PEFs are substantiated by the data, with evidence of informant triangulation (Malterud, 2001a; Miles et al., 2014). Logically, case 4 lacks references to a train operator because this case involved a bus connection operated by the airline.

Analysis of the first column highlights variation in how strongly categories of support processes and external conditions are rooted in the data. While most categories were consistently mentioned, *'IT'*, *'scheduling'*, *'market dynamics'*, and *'regulations'* appeared less frequently. These categories also exhibit fewer connections to PEFs in the subsequent columns.

A closer examination of the relationships between the categories and PEFs reveals several key patterns. First, *'IT'* primarily influences PEFs related to information provision and sharing, such as *'integrated booking'*, *'whole journey guidance'*, and *'disruption management'*.

Second, the category *'scheduling'* has limited impact, appearing only in connection to *'integrated booking'* (determining whether a journey is possible) and *'baggage management'* (determining whether integrated baggage is feasible).

Third, the *'(inter)organizational'* category affect all PEFs, underscoring the importance of both internal and external collaboration in developing multimodal air journeys.

Next, *'market dynamics'* impact only *'integrated booking'*, as its underlying elements (such as competition) largely determine whether an integrated journey is offered.

Furthermore, the category *'regulations'* influence the PEFs *'integrated booking'* and *'disruption management'*, and to a lesser extent, *'baggage management'*, as airline regulations require baggage to remain in controlled environments.

Finally, visualizing the connections between PEFs and support processes provides insight into which findings primarily apply to multimodal air journeys and which pertain mainly to air&rail (C1, C2, C3) or air&bus (C4) cases.

In case 4, integrated booking was mentioned less frequently because it was entirely managed by the airline. The connections with journey guidance were largely consistent across all four cases, suggesting that this aspect is not dependent on the mode of transport but is inherently important in any multimodal air journey. The importance of transfers appeared to be lower than in the air&rail cases. Additionally, baggage issues were less significant in the air&bus case and showed no connection to infrastructure, a link that was evident in the air&rail cases. Moreover, disruption management had minimal to no impact at the strategic, interorganizational, and regulatory levels in the air&bus case, which can again be attributed to the fact that the bus was entirely managed by the airline. Ultimately, the air&bus outlier case highlighted that infrastructure played a less critical role compared to the air&rail case.

The strongest relationships – those mentioned by the highest number of informants and cases – are observed between *'commercial'* and *'integrated booking'*, as well as between *'strategy'* and *'integrated booking'*. Following these, strong connections also emerge between *'market dynamics'* and *'integrated booking'*, and between *'(inter)organizational'* and *'whole journey guidance'*.

5. Discussion

Our findings indicate that practitioners in multimodal air transport considered five elements essential for a passenger-oriented journey: 1) integrated booking, 2) whole-journey guidance, 3) transfer time and ease, 4) baggage management, and 5) disruption management. These practitioner-identified priorities closely align with established research on the passenger experience.

Consistent with previous studies that have shown that integrated ticketing reduces perceived complexity (Babić et al., 2022; Román & Martín, 2014), our findings highlight that, according to practitioners, the primary benefit of simplified booking is the elimination of searches across multiple websites. Improving the booking process aligns with the EU's multimodal digital mobility services (MDMS) ambition to enable seamless cross-modal booking and ticketing, thereby enhancing user experience and interoperability.

Similarly, our results support earlier research indicating clear pre-trip and real-time information enhances passengers' sense of control and reduces perceived complexity (Allen et al., 2020; Antwi et al., 2020; Cascajo et al., 2019; Lubbe et al., 2011; Watkins et al., 2011). We expand this understanding by showing that practitioners noted passengers may not realize their journey is multimodal, suggesting that journey guidance should be continuous across legs and explicitly address transfers, rights, and responsibilities.

In line with prior studies that identify the transfer as a critical juncture (Babić et al., 2022; Chauhan et al., 2021; Durand & Romijn, 2023), our data underscore the importance of both the transfer time and ease. We found that airport-like amenities on the rail side could enhance the perceived quality of multimodal travel. While the co-location of modes is a prerequisite for transfers (Li et al., 2018), it was noted that this is insufficient without timetable coordination and integrated processes, such as check-in and baggage handling.

Regarding baggage management, previous research has demonstrated that integrated baggage systems reduce passenger effort (Babić

et al., 2024; Román & Martín, 2014; Wang et al., 2020). However, practitioners in our study agreed with that point, however they viewed full air-to-air style integration as largely unfeasible. Instead, they emphasized pragmatic solutions, such as third-party baggage services, enhanced luggage racks on trains, faster handling, and real-time updates.

Finally, consistent with existing literature that identifies disruption management as key to maintaining passenger trust through timely communication (Antwi et al., 2020; Babić et al., 2024; Cascajo et al., 2019), our findings reinforce this understanding by showing that practitioners believe passengers value assurance over speed.

While multimodal air travel inherently requires collaboration between transport systems (Babić et al., 2022; Toet et al., 2022, Toet et al., 2023), the airline representatives we interviewed tend to seek orchestration – or even control – over the passenger experience. As they sell tickets through their channels, they feel responsible for ensuring that passenger standards remain consistent with their disruption policies, baggage management, and in-flight expectations.

Our data also indicate that practitioners perceive passengers as adjusting their expectations based on their primary mode of transport, specifically, the airplane within multimodal air journeys. This aligns with existing research that identifies air travel as the central mode in multimodal air travel (Babić et al., 2022). As a result, in the cases we studied, the air leg was what we would call the 'baseline modality', the dominant mode in an integrated multimodal air journey. According to the informants, this baseline modality influences the passenger experience not only during transfers but also throughout the entire journey and sets passenger expectations. We found that these expectations include multiple aspects of the travel experience, including ticket booking, information provision, baggage handling, and passenger management during disruptions.

By uncovering the underlying mechanisms of delivering these PEFs to the passenger, we provide new insights into the intrinsic interdependence of services and physical infrastructure in multimodal air journeys. For instance, transfer services – including baggage handling, multimodal touchpoints, and priority lanes – rely on supporting infrastructure to function effectively. Similarly, while communication to the passenger is crucial, physical infrastructure plays a key role in its provision, such as through wayfinding. These insights closely align with existing literature on the role of service and physical infrastructure in multimodal air travel (Bell, 2019; Chauhan et al., 2021; Monzón et al., 2016; Toet et al., 2022). However, our findings show that this interdependence extends beyond infrastructure. We identified additional support process categories, including IT, scheduling, operations, commercial, (inter)organizational aspects, strategy, as well as broader external conditions such as market dynamics and regulations. All these categories are captured in underlying mechanisms and interconnected in the delivery of PEFs for multimodal air travel. Across these mechanisms we observed several recurring patterns that influence the overall delivery of multimodal air journeys.

Our findings indicate that the delivery of PEFs is hindered by the historical legacies of transport operators. While airlines primarily have a global focus, rail operators emerged from a national orientation (Bairras & Ardaiz, 2025; Vassallo & Fagan, 2007). Our research highlights the complexities arising from this disparity, demonstrating that the difference in focus presents a fundamental challenge for the delivery of multimodal air journeys, as the operational models and networks do not align with one another. We found that the long-standing operational histories of these operators predate the concept of multimodal air travel, meaning they are not designed or equipped to integrate with other transport systems. Working towards multimodality involves adapting physical infrastructure, such as station and airport facilities or the layout of transport vehicles like trains. However, adapting rail systems' infrastructure is costly, time-intensive, and can impact existing transport networks (Doomernik, 2015). In line with Toet et al. (2023), we found that determining who should bear these costs is particularly problematic, as we gained further insight that the involved actors have little financial incentive due to the relatively low number of multimodal air travelers (Román & Martín, 2014). We found that this may be attributed to many passengers being unaware that multimodal air travel exists or that such journeys can be booked as a single itinerary.

Even though informants noted that airlines have substantial experience collaborating with various stakeholders, we found that this primarily occurs within the same system (e.g., inter-airline agreements). Cross-system collaboration, as required for multimodal air travel, presents challenges such as adapting standardized IT systems. This is especially important for data sharing, as highlighted in previous research (Babić et al., 2022, Babić et al., 2024; Finger et al., 2019). Our findings agree that effective information exchange between transport systems is critical for guiding passengers throughout their journey and ensuring that transport operators have accurate passenger data. We observed that data sharing enables operators to respond appropriately and provide necessary guidance in case of disruptions. We extend this knowledge by demonstrating that, in practice, there is no clear ownership of multimodal air journeys. While existing research has highlighted the issue of responsibility sharing (Babić et al., 2022; European Parliament and of the Council, 2024), we deepen this understanding by discovering that the airlines involved in this study assume responsibility for the entire journey.

The issue of revenue sharing, as discussed by Babić et al. (2022) and Finger et al. (2019), was not observed in our study. This is primarily because airlines currently pay rail operators a fixed fee to enable ticket sales. As a result, there is no dual-provider model from the passenger's perspective, as the ticket is simply purchased through the airline. However, we found that if future regulations allow passengers to buy tickets independently of a dominant provider, revenue sharing may become a relevant issue.

We did identify another revenue-related challenge: the discrepancy in ticket fares between different transport systems. Airlines utilize yield optimization based on entire journeys and enforce boarding checks to prevent fare abuse and protect revenue integrity. This practice is not typically applied in rail transport. We found that this discrepancy makes it less attractive for airlines to offer combined tickets, as it creates opportunities for passengers to exploit differences in airline pricing structures. Our findings show that existing solutions, such as multimodal touchpoints at departure stations, require additional staff and infrastructure while also adding an extra step for passengers, making the journey more fragmented and operationally costly.

Returning to the idea that integrating more buses and trains with air travel can enhance sustainability (Baumeister, 2019; Eurocities, 2024; European Union Agency for Railways, 2024), our findings suggest a more nuanced reality. Driven by competition and capacity constraints in the airline industry, airlines tend to protect their airport slots, meaning they may continue operating the same routes even when a rail alternative is available. Literature further suggests that this can, in some cases, lead to increased emissions when airlines repurpose these freed-up slots for long-distance flights instead (Berveling et al., 2020; RLI, 2020). Our findings indicate that addressing this issue effectively requires regulatory intervention.

Although current regulations often constrain the delivery of key PEFs in multimodal air journeys – such as due to differing rules for disruption management among providers and strict baggage handling regulations – a new regulatory framework could instead facilitate progress. Such a framework can help align incentives, clarify responsibilities (for instance, regarding transfer times and ease), and ensure that data and ticketing systems are interoperable across modes. Policy measures should encourage operators to collaborate beyond their own systems and prioritize the passenger experience – not only by recognizing the importance of PEFs but also by placing the passenger at the center of delivering multimodal air journeys.

The study has several limitations that highlight opportunities for further exploration. First, it focuses exclusively on Europe, which limits

the global applicability of the findings. Although the study does not aim to generate universal knowledge due to its qualitative nature (Miles et al., 2014), we encourage readers to reflect on the conclusions drawn from data collected at hub airports in Europe that handled over 20 million passengers in 2019, where EU regulations apply. The findings should not readily be generalized beyond comparable multimodal air transport settings – or at least not without careful consideration and interpretation of contextual factors – but we invite future research to investigate cases outside Europe, taking into account the distinct contexts and conditions in other regions.

Second, although the set of interviewees was limited to twelve practitioners involved in four advanced cases, this scope inevitably restricts the generalizability of our findings. The cases were selected because they represent the most advanced examples of multimodal transport in Europe at the time of the research, but they cannot be assumed to reflect the full diversity of industry practice. In particular, the perspectives of airlines, rail operators, and bus companies not currently engaged in multimodal integration were not included, which limits insight into the reasons why broader adoption has been limited. Future studies including such perspectives across a wider range of contexts are necessary to provide more comprehensive and generalizable insights.

We included one outlier case (Miles et al., 2014) in this study that deviated from the air&rail connection, involving an air&bus connection fully operated by the airline. Although this resulted in fewer informants for the case, it provided valuable insights into what might have been specific for air&rail and which is more generalizable. Future research could explore a larger number of air&bus connections to build on these findings by examining services such as those offered by American Airlines on the U.S. East Coast.

The absence of direct input from bus operators also limits the study. Although the airline operated the bus leg in this case, including bus operators in future research could provide a more comprehensive understanding of operational and coordination challenges in multimodal transport.

Although additional connections between support processes and passenger experience factors may be conceivable, Fig. 4 reflects only those relationships that emerged directly from the empirical data, specifically in response to the research question: *"What influences the effective delivery of these passenger experience factors to travelers in Europe?"* This study has therefore been deliberately limited to addressing this question. Future research could explore whether support processes also influence other factors or aspects beyond those identified in this study.

Furthermore, the concept of baseline modality emerged from the data, as informants explicitly mentioned it. However, this interpretation comes from practitioners' perspectives rather than those of passengers. Future research could explore whether baseline modality resonates with travelers, influences their expectations, or exists solely among practitioners. It would also be valuable to explore whether the concept of baseline modality applies to other service contexts, such as MaaS. The dynamics of multimodal services may vary depending on which modality is dominant, or whether a more balanced distribution of influence exists.

Additionally, collaboration emerged as a recurring theme and is critical for the delivery of multimodal air transport. Future research could delve deeper into how collaboration can be improved in practice, drawing from real-world examples and case studies.

More general, this study shows that practitioners, as creators of multimodal travel services, shape the passenger experience primarily based on organizational rationale rather than grounded in passenger-driven insights. While they demonstrate an understanding of passenger needs through identified PEFs their priorities are often influenced by operational and institutional considerations. Future studies that actively involve passengers – such as through co-creation workshops or experimental interventions – could offer a complementary perspective and help bridge the gap between practitioners' beliefs about what passengers value and what passengers experience. Additionally, further research could explore how passenger needs can be the guiding force in the service design and delivery process of multimodal air journeys.

6. Conclusion

Based on interviews with multimodal air transport practitioners, this study identifies five key PEFs for passenger-oriented multimodal air journeys: 1) integrated booking, 2) whole journey guidance, 3) transfer time and ease, 4) baggage management, and 5) disruption management.

Our findings reveal that delivering these PEFs requires more than just infrastructure and service integration across travel modes. A range of support processes within and between organizations must also align, such as IT, scheduling, operations, commercial, (inter)organizational, and strategy. In addition, external conditions – market dynamics and regulatory frameworks – play a pivotal role in either enabling or constraining these efforts.

We discovered that these underlying support processes converge in delivery mechanisms that influence the delivery of PEFs for multimodal air travel. By surfacing the interplay between PEFs, support processes, and external conditions, this study contributes to the literature on multimodal transport and the emerging discourse on passenger experiences. The findings provide valuable insights for future research to enhance the understanding of how support processes and external conditions manifest in real-world projects and explore potential solutions to effectively advance passenger-oriented multimodal air journeys.

In conclusion, while airlines often take the lead in shaping the multimodal experience, progress remains constrained by fragmented responsibilities, misaligned incentives among operators, and market competition. Delivering truly passenger-oriented multimodal air journeys requires not only effective cross-system collaboration but also a regulatory framework that creates the right conditions for integration and sustainability. Policymakers should ensure that governance structures and incentives encourage cooperation across transport modes. Equally, operators must design and manage multimodal air transport with the passenger in mind, prioritizing seamlessness over the protection of individual systems or market positions.

Declaration of generative AI in scientific writing

During the preparation of this work the first author used EditGPT for readability and grammar checks. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

CRediT authorship contribution statement

Aniek Toet: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jasper van Kuijk:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Klaas Boersma:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Sicco Santema:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships that may be considered potential competing interests: the first author of the study, Aniek Toet, reports receiving financial support from Royal Schiphol Group and performing her research as an embedded researcher in the organization, and the third author, Klaas Boersma, is employed by Royal Schiphol Group.

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Data availability

Due to the data collected including personal elements and judgments that can be traced back to the researchers and practitioners to whom the transcripts belong, this study treats the data as confidential. However, the data are available from the corresponding author upon reasonable request.

References

Aleta, A., Meloni, S., & Moreno, Y. (2017). A multilayer perspective for the analysis of urban transportation systems. *Scientific Reports, 7*(1), Article 44359. https://doi.org/10.1038/srep44359

Allen, J., Bellizzi, M. G., Eboli, L., Forciniti, C., & Mazzulla, G. (2020). Latent factors on the assessment of service quality in an Italian peripheral airport. *Transportation Research Procedia, 47*, 91–98. https://doi.org/10.1016/j.trpro.2020.03.083

Antwi, C. O., Fan, C., Ihnatushchenko, N., Aboagye, M. O., & Xu, H. (2020). Does the nature of airport terminal service activities matter? Processing and non-processing service quality, passenger affective image and satisfaction. *Journal of Air Transport Management, 89*, Article 101869. https://doi.org/10.1016/j.jairtraman.2020.101869

Babić, D., Colovic, A., Dožić, S., Kalić, M., Simić, T. K., Kukić, K., … Pilone, S. G. (2024). How to build a more sustainable passenger air transport system: Multimodal experience. *Transportation Engineering, 16*, Article 100245. https://doi.org/10.1016/j.treng.2024.100245

Babić, D., Kalić, M., Janić, M., Dožić, S., & Kukić, K. (2022). Integrated door-to-door transport Services for air Passengers: From Intermodality to multimodality. *Sustainability, 14*(11), 6503. https://doi.org/10.3390/su14116503

Bairras, P., & Ardaiz, I. A. (2025). The slow and difficult implementation of high speed rail interoperability in Europe: The case of the Atlantic corridor. *Transportation Research Procedia, 82*, 2546–2558. https://doi.org/10.1016/j.trpro.2024.12.204

Baumeister, S. (2019). Replacing short-haul flights with land-based transportation modes to reduce greenhouse gas emissions: The case of Finland. *Journal of Cleaner Production, 225*, 262–269. https://doi.org/10.1016/j.jclepro.2019.03.329

Bell, D. (2019). Intermodal mobility hubs and user needs. *Social Sciences, 8*(2), 65. https://doi.org/10.3390/socsci8020065

Bitner, M. J., Ostrom, A. L., & Morgan, F. N. (2008). Service blueprinting: A practical technique for service innovation. *California Management Review, 50*(3), 66–94. https://doi.org/10.2307/41166446

Braun, V., & Clarke, V. (2012). Thematic analysis. In H. Cooper, P. M. Camic, D. L. Long, A. T. Panter, D. Rindskopf, & K. J. Sher (Eds.), *APA handbook of research methods in psychology, vol 2: Research designs: Quantitative, qualitative, neuropsychological, and biological* (pp. 57–71). American Psychological Association. https://doi.org/10.1037/13620-004.

Camacho, T., Foth, M., Rakotonirainy, A., Rittenbruch, M., & Bunker, J. (2016). The role of passenger-centric innovation in the future of public transport. *Public Transport, 8*(3), 453–475. https://doi.org/10.1007/s12469-016-0148-5

Canale, A., Tesoriere, G., & Campisi, T. (2019). *The MAAS development as a mobility solution based on the individual needs of transport users.*, 160005. https://doi.org/10.1063/1.5138073

Cascajo, R., Lopez, E., Herrero, F., & Monzon, A. (2019). User perception of transfers in multimodal urban trips: A qualitative study. *International Journal of Sustainable Transportation, 13*(6), 393–406. https://doi.org/10.1080/15568318.2018.1476632

Chakrabarti, A. (2019). *Research into Design for a Connected World: Proceedings of ICoRD 2019 volume 2 (vol. 135)*. Springer Singapore. https://doi.org/10.1007/978-981-13-5977-4

Chauhan, V., Gupta, A., & Parida, M. (2021). Demystifying service quality of multimodal transportation hub (MMTH) through measuring users' satisfaction of public transport. *Transport Policy, 102*, 47–60. https://doi.org/10.1016/j.tranpol.2021.01.004

Chiambaretto, P., & Decker, C. (2012). Air–rail intermodal agreements: Balancing the competition and environmental effects. *Journal of Air Transport Management, 23*, 36–40. https://doi.org/10.1016/j.jairtraman.2012.01.012

Docherty, I., Marsden, G., & Anable, J. (2018). The governance of smart mobility. *Transportation Research Part A: Policy and Practice, 115*, 114–125. https://doi.org/10.1016/j.tra.2017.09.012

Doomernik, J. E. (2015). Performance and efficiency of high-speed rail systems. *Transportation Research Procedia, 8*, 136–144. https://doi.org/10.1016/j.trpro.2015.06.049

Döringer, S. (2021). 'The problem-centred expert interview'. Combining qualitative interviewing approaches for investigating implicit expert knowledge. *International Journal of Social Research Methodology, 24*(3), 265–278. https://doi.org/10.1080/13645579.2020.1766777

Durand, A., & Romijn, G. (2023). Substitutiemogelijkheden van luchtvaart naar spoor in 2030 en 2040 (report). *Kennisinstituut voor Mobiliteitsbeleid (KiM)*.

Eboli, L., & Mazzulla, G. (2015). Relationships between rail passengers' satisfaction and service quality: A framework for identifying key service factors. *Public Transport, 7*(2), 185–201. https://doi.org/10.1007/s12469-014-0096-x

Eisenhardt, K. M. (1989). Building theories from case study research. *Academic Management Review, 14*, 532–550.

Eurocities. (2024). *Policy statement on prioritising fast and high-quality passenger train connection in Europe*. https://eurocities.eu/wp-content/uploads/2024/10/FINAL-Policy-Statement-on-Prioritising-fast-and-high-quality-passenger-train-connections-in-Europe.pdf.

European Commission. (2011). *Roadmap to a Single European Transport Area – Towards a competitive and resource efficient transport system. No. 144*. https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52011DC0144.

European Commission. (2021). *Sustainable & Smart Mobility Strategy—Putting European Stransport on track for the future*.

European Parliament and of the Council. (2024). *Passenger rights in the context of multimodal journeys*. https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/767200/EPRS_BRI(2024)767200_EN.pdf.

European Union Agency for Railways. (2024). Rail environmental report. *Publications Office.*. https://doi.org/10.2821/243881

Finger, M., Montero, J., & Serafimova, T. (2019). *Towards EU-wide multimodal ticketing and payment systems*.

Geng, Y., Yu, J., Lin, B., Wang, Z., & Huang, Y. (2017). Impact of individual IEQ factors on passengers' overall satisfaction in Chinese airport terminals. *Building and Environment, 112*, 241–249. https://doi.org/10.1016/j.buildenv.2016.11.040

Geurs, K. T., La Paix, L., & Van Weperen, S. (2016). A multi-modal network approach to model public transport accessibility impacts of bicycle-train integration policies. *European Transport Research Review, 8*(4), 25. https://doi.org/10.1007/s12544-016-0212-x

Givoni, M., & Banister, D. (2006). Airline and railway integration. *Transport Policy, 13*(5), 386–397. https://doi.org/10.1016/j.tranpol.2006.02.001

Guidon, S., Wicki, M., Bernauer, T., & Axhausen, K. (2020). Transportation service bundling – Fo r whose benefit? Consumer valuation of pure bundling in the passenger transportation market. *Transportation Research Part A: Policy and Practice, 131*, 91–106. https://doi.org/10.1016/j.tra.2019.09.023

Hennink, M., & Kaiser, B. N. (2022). Sample sizes for saturation in qualitative research: A systematic review of empirical tests. *Social Science & Medicine, 292*, Article 114523. https://doi.org/10.1016/j.socscimed.2021.114523

Hernandez Bueno, A. V. (2021). Becoming a passenger: Exploring the situational passenger experience and airport design in the Copenhagen airport. *Mobilities, 16*(3), 440–459. https://doi.org/10.1080/17450101.2020.1864114

Hernandez, S., & Monzon, A. (2016). Key factors for defining an efficient urban transport interchange: Users' perceptions. *Cities, 50*, 158–167. https://doi.org/10.1016/j.cities.2015.09.009

Hickman, R., Chen, C.-L., Chow, A., & Saxena, S. (2015). Improving interchanges in China: The experiential phenomenon. *Journal of Transport Geography, 42*, 175–186. https://doi.org/10.1016/j.jtrangeo.2014.12.004

Janic, M. (2011). Assessing some social and environmental effects of transforming an airport into a real multimodal transport node. *Transportation Research Part D: Transport and Environment, 16*(2), 137–149. https://doi.org/10.1016/j.trd.2010.10.002

Jiang, F., Wang, L., & Huang, S. (2022). Analysis of the transfer time and influencing factors of air-rail integration passengers: A case study of Shijiazhuang Zhengding international airport. *Sustainability, 14*(23), Article 16193. https://doi.org/10.3390/su142316193

Kivela, M., Arenas, A., Barthelemy, M., Gleeson, J. P., Moreno, Y., & Porter, M. A. (2014). Multilayer networks. *Journal of Complex Networks, 2*(3), 203–271. https://doi.org/10.1093/comnet/cnu016

Li, X., Jiang, C., Wang, K., & Ma, J. (2018). Determinants of partnership levels in air-rail cooperation. *Journal of Air Transport Management, 71*, 88–96. https://doi.org/10.1016/j.jairtraman.2018.06.002

Lubbe, B., Douglas, A., & Zambellis, J. (2011). An application of the airport service quality model in South Africa. *Journal of Air Transport Management, 17*(4), 224–227. https://doi.org/10.1016/j.jairtraman.2010.08.001

Mager, B. (2009). Service design as an emerging field. *Designing services with innovative. Methods, 1*.

Magnani, G., & Gioia, D. (2023). Using the Gioia methodology in international business and entrepreneurship research. *International Business Review, 32*(2), Article 102097. https://doi.org/10.1016/j.ibusrev.2022.102097

Malterud, K. (2001a). Qualitative research: Standards, challenges, and guidelines. *The Lancet, 358*(9280), 483–488. https://doi.org/10.1016/S0140-6736(01)05627-6

Malterud, K. (2001b). The art and science of clinical knowledge: Evidence beyond measures and numbers. *The Lancet, 358*(9279), 397–400. https://doi.org/10.1016/S0140-6736(01)05548-9

Marquez, V. (2019). *Landside| airside: Why airports are the way they are*. Springer.

Meroni, A., & Sangiorgi, D.. (2016). *Design for services*. Routledge.

Miles, M., Huberman, M., & Saldaña, J. (2014). *Qualitative data analysis: A methods sourcebook* (3rd ed.). Sage Publications.

Monzón, A., Hernández, S., & Di Ciommo, F. (2016). Efficient urban interchanges: The City-HUB model. *Transportation Research Procedia, 14*, 1124–1133. https://doi.org/10.1016/j.trpro.2016.05.183

Nam, S., Ha, C., & Lee, H. C. (2018). Redesigning in-flight service with service blueprint based on text analysis. *Sustainability, 10*(12), 4492. https://doi.org/10.3390/su10124492

Nash, C. (2010). European rail reform and passenger services – The next steps. *Research in Transportation Economics, 29*(1), 204–211. https://doi.org/10.1016/j.retrec.2010.07.025

Nechyporuk, M., Pavlikov, V., & Kritskiy, D. (Eds.). (2022), *Vol. 367. Integrated Computer Technologies in Mechanical Engineering - 2021: Synergetic engineering*. Springer International Publishing. https://doi.org/10.1007/978-3-030-94259-5.

Nielsen, O. A., Eltved, M., Anderson, M. K., & Prato, C. G. (2021). Relevance of detailed transfer attributes in large-scale multimodal route choice models for metropolitan public transport passengers. *Transportation Research Part A: Policy and Practice, 147*, 76–92. https://doi.org/10.1016/j.tra.2021.02.010

Papastathopoulou, P., & Hultink, E. J. (2012). New service development: An analysis of 27 years of research *. *Journal of Product Innovation Management, 29*(5), 705–714. https://doi.org/10.1111/j.1540-5885.2012.00944.x

Patton, M. Q. (2015). *Qualitative research & evaluation methods: Integrating theory and practice* (4th ed.). SAGE.

Regulation (EU) 2024/1679 on Union Guidelines for the Development of the Trans-European Transport Network, OJ L 2024/1679, 19.6.2024. https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ:L_202401679, (2024).

RLI. (2020). Verzet de wissel: Naar beter internationaal reizigersvervoer per trein (report). *Raad voor de Leefomgeving en Infrastructuur (Rli)*.

Rodrigue, J.-P. (2024). *The geography of transport systems* (6th ed.). Routledge.

Román, C., & Martín, J. C. (2014). Integration of HSR and air transport: Understanding passengers' preferences. *Transportation Research Part E: Logistics and Transportation Review, 71*, 129–141. https://doi.org/10.1016/j.tre.2014.09.001

Rosenbaum, M. S., Otalora, M. L., & Ramírez, G. C. (2017). How to create a realistic customer journey map. *Business Horizons, 60*(1), 143–150. https://doi.org/10.1016/j.bushor.2016.09.010

Schmitt, D., & Gollnick, V. (2016). Air Transport System. *Springer Vienna.*. https://doi.org/10.1007/978-3-7091-1880-1

Shostack, G. L. (1984). Designing services that deliver. *Harvard Business Review, 84115*.

Silverman, D. (2009). *Doing qualitative research* (3rd ed.). SAGE Publications.

Song, F., Hess, S., & Dekker, T. (2018). Accounting for the impact of variety-seeking: Theory and application to HSR-air intermodality in China. *Journal of Air Transport Management, 69*, 99–111. https://doi.org/10.1016/j.jairtraman.2018.02.008

Toering, A., De Bruijne, M., & Veeneman, W. (2025). Exploring governance challenges of sustainable infrastructure development on the nexus between energy and mobility. *Npj Sustainable Mobility and Transport, 2*(1), 1. https://doi.org/10.1038/s44333-024-00018-0

Berveling, J., Zijlstra, T., Knoope, M., & Huibregtse, O. (2020). *Op de Groene Tour: de bijdrage van gedragsinterventies aan het verduurzamen van de luchtvaart (report). Kennisinstituut voor Mobiliteitsbeleid (KiM)*.

Toet, A., Van Kuijk, J., Boersma, K., & Santema, S. (2023, October 9). *Themes for an airport hub in the transition towards a multimodal transport hub – An embedded researcher's perspective*. IASDR 2023: Life-changing design, Milan, Italy. Doi:10.21606/iasdr.2023.272.

Toet, A., van Kuijk, J., & Santema, S. (2022). Transforming airport hubs into future-proof multimodal transport hubs. In *Relating systems thinking and design (RSD11) symposium: Possibilities and practices of systemic design, Brighton, UK*.

UIC. (2024). *High-speed rail Atlas*. International Union of Railways (UIC).

Vassallo, J. M., & Fagan, M. (2007). Nature or nurture: Why do railroads carry greater freight share in the United States than in Europe? *Transportation, 34*(2), 177–193. https://doi.org/10.1007/s11116-006-9103-7

Veeneman, W. (2023). Governance, COVID responses, and lessons on decision-making in uncertainty. In *Transportation amid pandemics* (pp. 441–450). Elsevier. https://doi.org/10.1016/B978-0-323-99770-6.00019-3.

Vuchic, V. R. (2007). *Urban transit systems and technology*. John Wiley & Sons.

Wan, Y., Ha, H.-K., Yoshida, Y., & Zhang, A. (2016). Airlines' reaction to high-speed rail entries: Empirical study of the northeast Asian market. *Transportation Research Part A: Policy and Practice, 94*, 532–557. https://doi.org/10.1016/j.tra.2016.10.014

Wang, B., Loo, B., & Li, L. (2020). Situating high-speed railway stations within local urban contexts: Passenger satisfaction with intermodal integration at the Hong Kong HSR station. *Built Environment, 46*(3), 362–378. https://doi.org/10.2148/benv.46.3.362

Wang, B., Shen, G., Wang, X., Dong, Y., & Li, Z. (2024). Hub-and-spoke network optimization with flow delay cost: The case of goods delivery on urban logistics networks in eastern China. *Mathematics, 12*(10), 1496. https://doi.org/10.3390/math12101496

Watkins, K. E., Ferris, B., Borning, A., Rutherford, G. S., & Layton, D. (2011). Where is my bus? Impact of mobile real-time information on the perceived and actual wait time of transit riders. *Transportation Research Part A: Policy and Practice, 45*(8), 839–848. https://doi.org/10.1016/j.tra.2011.06.010

Wattanacharoensil, W., Schuckert, M., & Graham, A. (2016). An airport experience framework from a tourism perspective. *Transport Reviews, 36*(3), 318–340. https://doi.org/10.1080/01441647.2015.1077287

Wu, Y., Morlotti, C., & Mantin, B. (2024). Shopping or dining? On passenger dwell time and non-aeronautical revenues. *Journal of Air Transport Management, 118*, 102620.

Xia, W., & Zhang, A. (2017). Air and high-speed rail transport integration on profits and and welfare: Effects of air-rail connecting time. *Journal of air transport management, 65*, 181–190.

Yang, H., & Zhang, A. (2012). Effects of high-speed rail and air transport competition on prices, profits and welfare. *Transportation Research Part B: Methodological, 46*(10), 1322–1333. https://doi.org/10.1016/j.trb.2012.09.001

Yin, R. K. (2018). *Case study research and applications: Design and methods* (6th ed.). SAGE Publications.

Zhou, T., Tan, R., & Sedlin, T. (2018). Planning modes for major transportation infrastructure projects (MTIPs): Comparing China and Germany. *Sustainability, 10*(10), 3401. https://doi.org/10.3390/su10103401